

Use case	Authentication	
Release (id)	UC-02, v1.0, 26 April 2026	
Context of Use	A user needs to register, log in, or log out of the system to access personalised or restricted features.	
Scope	Manage user identity: registration, login, logout, and session control.	
Level	Primary task	
Primary actor	General User (unregistered → registered; or registered user)	
StakeHolder & interest	StakeHolder	Interest
	General User	Wants a secure but simple login/registration process
	Railway administrator	Needs reliable authentication to protect data and comply with GDPR
	Login System (external API)	Must correctly verify credentials
Preconditions	→ The system is connected to the Login System API (NFR 24) → The Database is reachable for storing/retracting credentials → For registration: user provides valid email and password (NFR 16)	
Minimal Guarantees	→ No authentication action leaves the system in an inconsistent state (e.g., half-registered user) → Passwords are never stored in plain text (NFR 18)	
Success Guarantees	→ A new user account is created (FR 12) → A registered user successfully logs in (FR 28) → A logged-in user can log out, ending the session (FR 28)	
Trigger	The user clicks “Login”, “Register”, or “Logout” in the interface.	
Description (Main Success Scenario)	Step	Action
	1	User selects “Register” (FR 12), “Login” (FR 28), or “Logout” (FR 28).
	Registration branch:	
	2	User fills in name, surname, email, and password (NFR 16).
	3	System sends data to the Login System API; API checks email uniqueness.
	4	System creates the account and automatically logs the user in.
	Login branch:	
	2'	User enters email and password.
	3'	System validates credentials via the Login System API.
	4'	System starts a user session with a timeout of 30 minutes (NFR 15).
	Logout branch:	
	2''	User clicks “Logout”.
	3''	System terminates the session and redirects to the public homepage.
Extension	Step	Other action
	3a	If the email already exists during registration, system returns “Email already registered” (FR 12).
	3b	<i>If the password does not meet security requirements (NFR 16), system asks for a stronger password.</i>
	3a ₁	If incorrect credentials are provided, system increments failed attempt counter. After 5 failed attempts per IP in 1 minute, the account is temporarily locked (NFR 20).
	3b ₁	If the Login System API is unavailable, system shows “Service temporarily unavailable, try again later”.
Technology & data Variations	→ Authentication can be performed via the external Login System API using OAuth2 or similar (NFR 24). → On mobile, biometric login (fingerprint/face) may be supported if the device allows. → Session data is stored server-side; the client only holds a session token.	